What to expect in these two lectures

CV2090 Probability & Statistics for Civil Engineers · summary, lecture 1 of 2

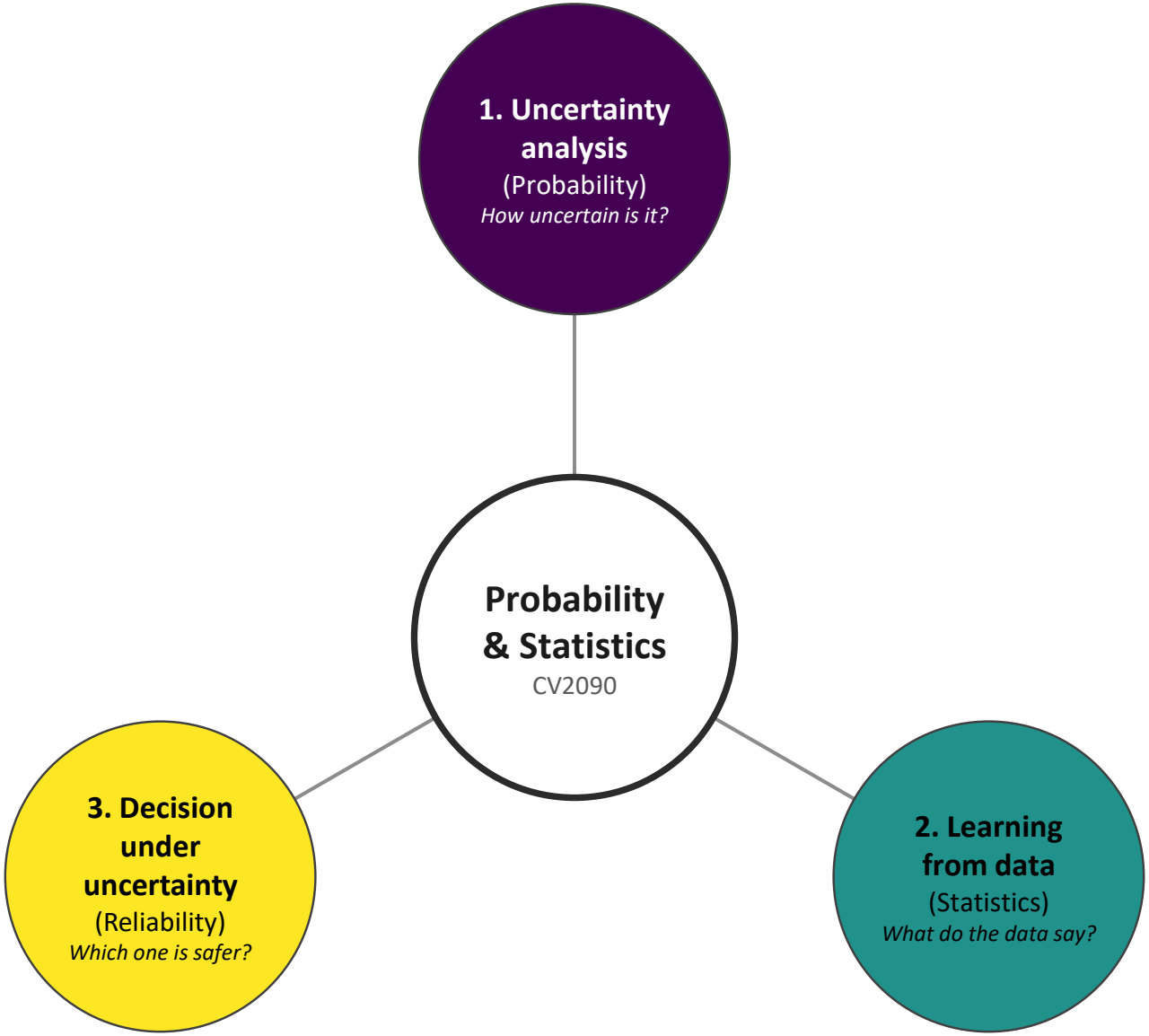
Lecture 1 (today)

- **The course map: all three branches at a glance**
- **1. Uncertainty analysis**
- **2. Learning from data**
- **For each branch: its parts and stories, which tool when, know this, traps the questions test**

Lecture 2

- 3. Decision under uncertainty: its parts and stories, which tool when, know this, traps the questions test
- Ten ideas to keep
- Beyond the course: how the parts connect, practice and communication, ten questions to ask of any claim
- Where to go next

Root question: *If "certainty does not exist", how should engineers reason?*



Why this course?

- Steel ball drop 100×: same physics, different t
- Five acts: real world → observations → **models** → prediction → engineering
- Three layers: what we have (physics/data, **past**) / what we don't know (true reality, **future**) / what we must decide (design bridge/diagnose, **present**)
- Priming poll: **Football (11) + US flag (13) + Phosphorus (17) → Random number** $\in [11, 20]$ "Let's ignore opinions and ask the data!"
- "All models are wrong, but some are useful." (George Box)

1. Uncertainty analysis
(Probability)
How uncertain is it?

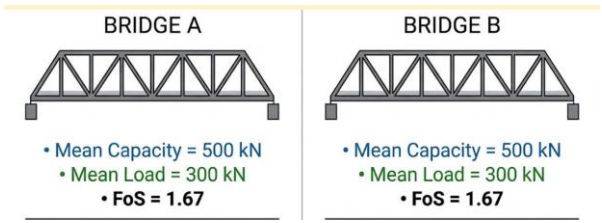
From FoS to probability

- Concrete cubes 13, 15, 17, 12, 13, 13, 20: which is the "true" strength?
- "Deterministic engineering is NOT wrong, it is incomplete!"
- Safety margin $Z = R - S$
- Two bridges, same FoS = 1.67: which one is safer?
- Probability × Consequence → Risk

Probability & Statistics
CV2090

3. Decision under uncertainty
(Reliability)
Which one is safer?

2. Learning from data
(Statistics)
What do the data say?

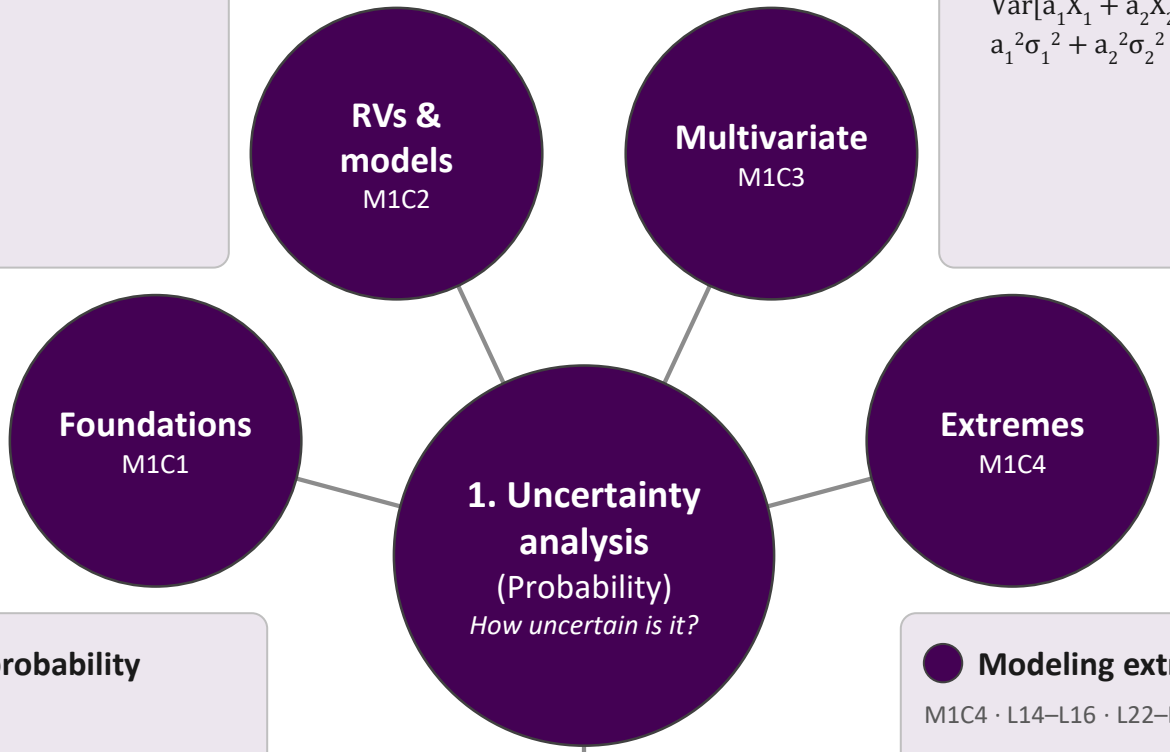


Bridge A vs. Bridge B

Root question: *If "certainty does not exist", how should engineers reason?*



Not covered in class.



RVs & probability models

M1C2 · L7–L13

- **Key equations:**
  $E[g(X)] = \int g(x)\, f_X(x)\, dx$
  $\sigma^2 = E[X^2] - \mu^2$

Multivariate probability

M1C3 · L17–L21

- **Named:** Ulam, von Neumann, Pythagoras (SRSS)
- **Key equation:**
  $Var[a_1X_1 + a_2X_2] = a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + 2a_1a_2Cov[X_1,X_2]$

Foundations of probability

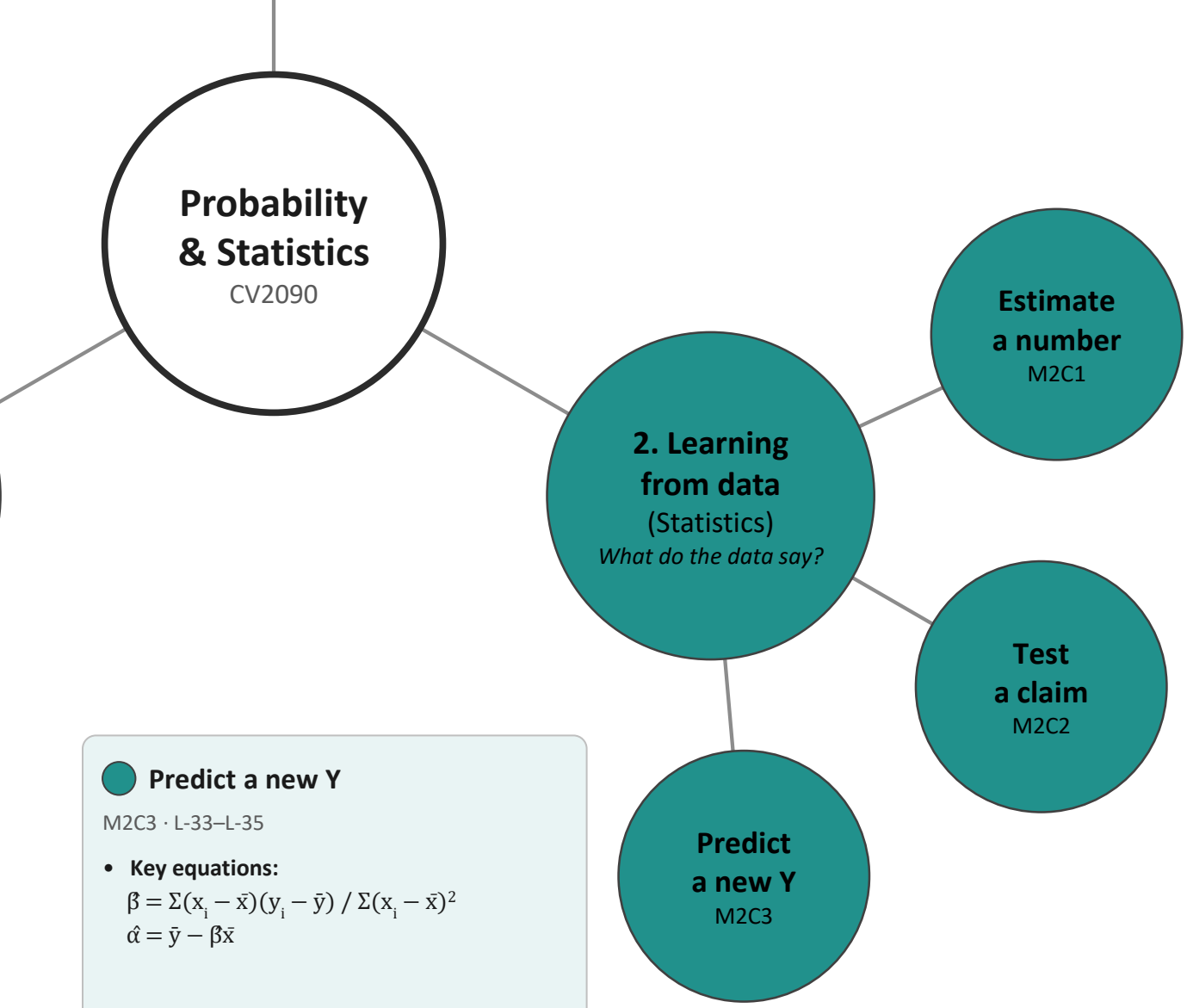
M1C1 · L3–L6

- **Named:** Bayes, Bertrand, De Morgan
- **Key equations:**
  $Pr(E_i|A) = Pr(A|E_i)Pr(E_i) \,/\, Pr(A)$
  $Pr(A) = \Sigma_j\, Pr(A|E_j)Pr(E_j)$

Modeling extremes

M1C4 · L14–L16 · L22–L25

- **Named:** de Moivre, Gauss, van Veen, Fisher–Tippett–Gnedenko
- **Key equations:**
  $F_{Yn}(y) = [F_X(y)]^n,\ T_r = 1/p_{f,1y}$



Estimate a number

M2C1 · L26–L27

- **Key equation:**
  $\bar{x} \pm z_{\alpha/2}\, \sigma/\sqrt{n}$

Test a claim

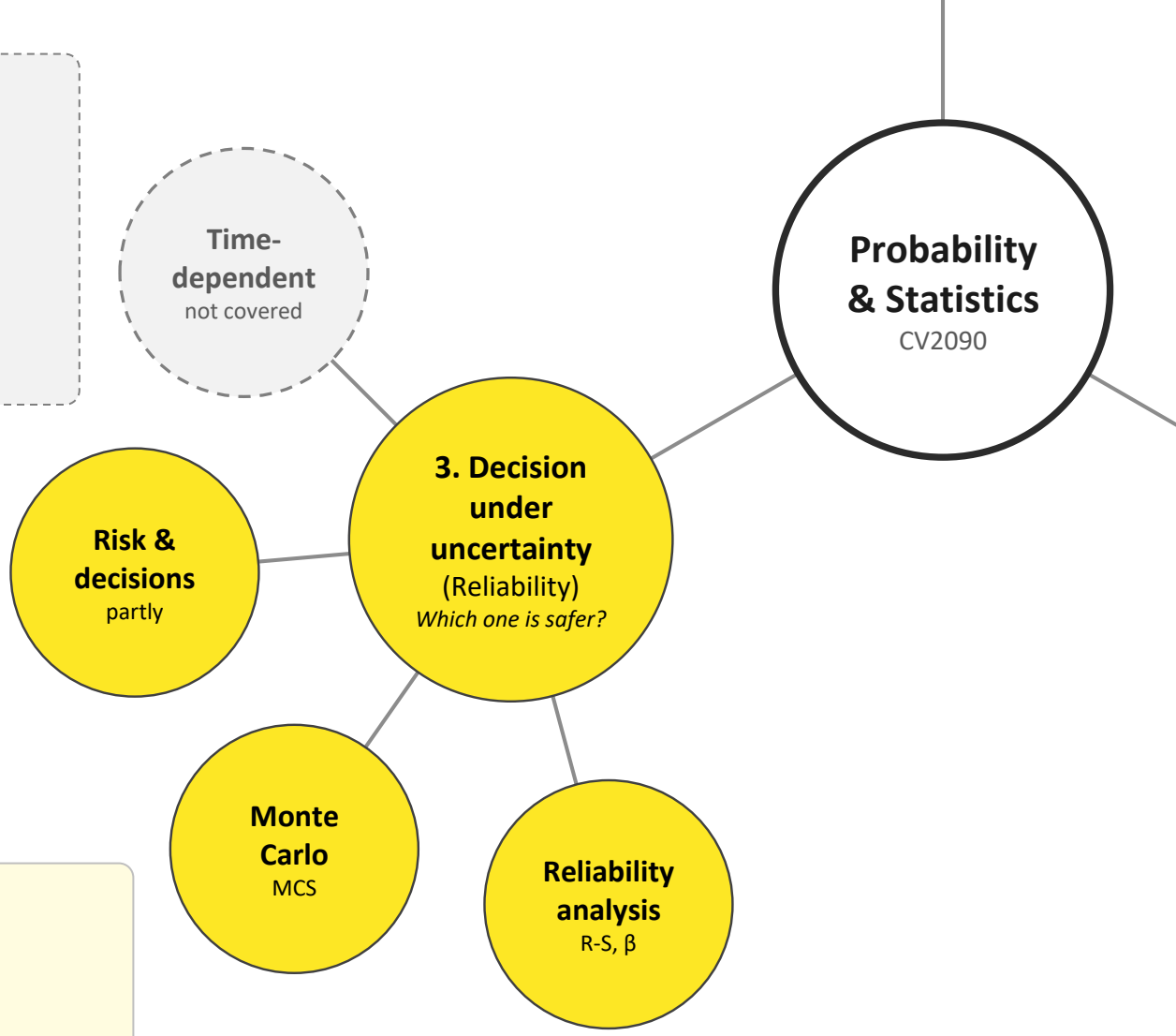
M2C2 · L-29–L-32

- **Key equations:**
  $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$
  $\text{power} = 1 - \beta$

Predict a new Y

M2C3 · L-33–L-35

- **Key equations:**
  $\hat{\beta} = \Sigma(x_i - \bar{x})(y_i - \bar{y}) \,/\, \Sigma(x_i - \bar{x})^2$
  $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$



Time-dependent reliability

not covered this term

- Related, taught in M1C4: Poisson model, return period, design life

Risk & decisions

partly covered · L-36

- **Key:** risk = expected loss; 10^{-3}: act now; 10^{-6}: too rare to act

Computational reliability

L-37

- **Named:** Ulam
- **Key equations:**

$$\hat{p}_f = (1/N)\,\Sigma\, q_i, \quad \delta \cong \sqrt{(1/(N\,p_F))}$$

Reliability analysis

L-36–L-38

- **Key equations:** $g(R,S) = R - S, \quad \beta = \mu_Z/\sigma_Z, \quad p_f \cong \Phi(-\beta)$

Root question: *If "certainty does not exist", how should engineers reason?*



Not covered this term.

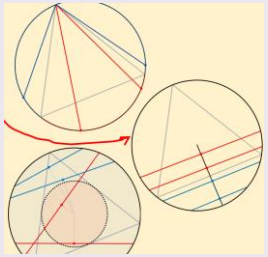
1. Uncertainty analysis (Probability): four parts

Question: *How uncertain is it?*

Foundations of probability

M1C1 · L3–L6

- Sample space, events, axioms; Venn, MECE, De Morgan
- Union, complement, inclusion–exclusion
- Conditional probability, independence, multiplication rule
- Total probability rule: “forward prediction”
- Bayes’ rule: “reverse prediction” (inference)

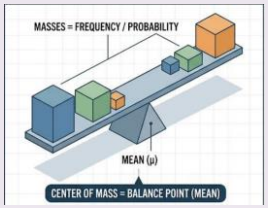


Stories: birthdays, Reyansh, Bertrand, O. J. Simpson

RVs & probability models

M1C2 · L7–L13

- rv: a function from S to $\mathbb{R}$; discrete vs. continuous
- PMF, PDF, CDF
- Mean, variance, moments, skewness, kurtosis
- Bernoulli, Binomial, Geometric, Poisson
- LOTUS; functions of one rv (CDF method)

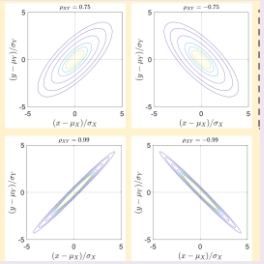


Stories: Greenshields, bridge bearings, Datasaurus

Multivariate probability

M1C3 · L17–L21

- Joint, marginal, conditional distributions
- Covariance, ρ_{XY}; $\rho = 0 \not\Rightarrow$ independence
- Variance of a sum; diversification
- Propagation: linear laws; Taylor approximations
- Aleatory vs. epistemic; Monte Carlo (brief)



Stories: two buildings, mutual funds, Ulam’s solitaire

Modeling extremes

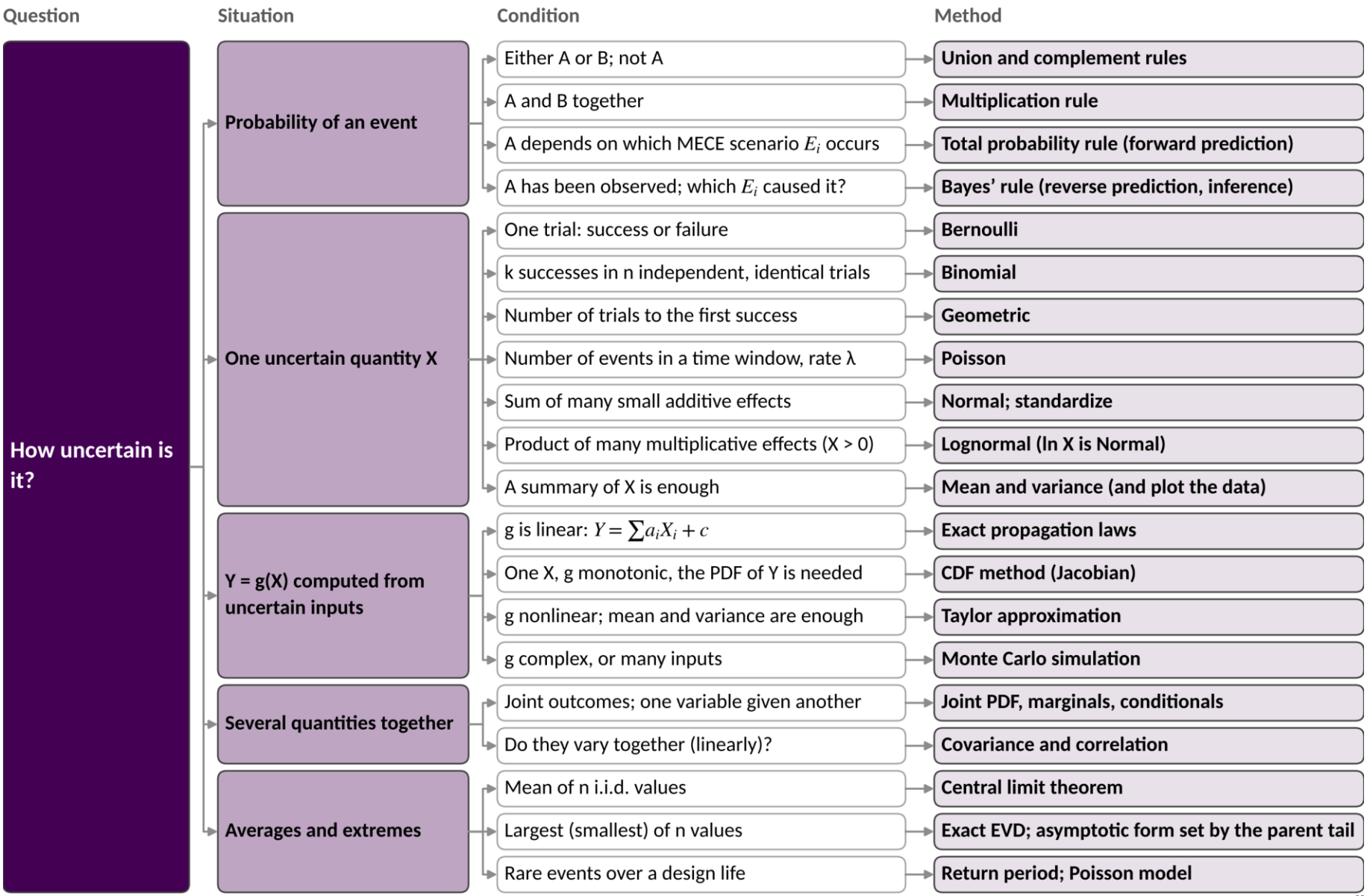
M1C4 · L14–L16 · L22–L25

- Normal (additive effects); $Z = (X - \mu)/\sigma$
- Lognormal (multiplicative effects); λ, ζ
- CLT: sample mean $\approx \mathcal{N}(\mu, \sigma^2/n)$
- Min and max of n rvs; Gumbel, Fréchet, Weibull; GEV
- Return period, design life, Poisson



Stories: Fukushima, 1953 North Sea flood, dike rings

Which tool when? 1. Uncertainty analysis



Know this: 1. Uncertainty analysis (1 of 2)

Foundations of probability (M1C1) · RVs & probability models (M1C2)

Know this

- **Bayes' rule** updates a prior with evidence: $\Pr(E_i|A) \propto \Pr(A|E_i)\Pr(E_i)$. The posterior always depends on the prior.
- **Total probability** is forward prediction ("what will happen?"); **Bayes** is reverse prediction ("what caused it?"). Both use MECE events.
- A **PDF** is a density: area under f_X is probability, f_X has no upper bound, and $\Pr(X = x) = 0$ for a continuous rv.
- **Mean** is the center of mass and **variance** the moment of inertia of the probability "mass": $\sigma^2 = E[X^2] - \mu^2$.

Don't confuse

- *"Pr(guilty | DNA match) = Pr(DNA match | guilty)."*
  → They differ. The jury needs Pr(guilty | DNA match), which also depends on the prior.
- *"Reyansh is more likely a Tesla engineer and a policy buff, because that fits his story."*
  → $\Pr(A \cap B) \leq \Pr(A)$: "Probability, not plausibility."
- *"Equal μ and σ^2 mean the same distribution."*
  → They do not identify a distribution (Datasaurus Dozen). Eyeball the data.
- *"$E[X^2] = (E[X])^2$."*
  → $E[g(X)] = g(E[X])$ only for affine g; otherwise use LOTUS.

Know this: 1. Uncertainty analysis (2 of 2)

Multivariate probability (M1C3) · Modeling extremes (M1C4)

Know this

- Knowing each rv separately is not enough; the **joint PDF** holds full information, and marginals and conditionals follow from it.
- Correlation changes the spread of a sum:
  $\mathrm{Var}(D + L) = \sigma_D^2 + \sigma_L^2 + 2\rho_{DL}\sigma_D\sigma_L$.
  With $\rho < 1$, diversification reduces risk.
- **Additive** effects lead to the Normal; **multiplicative** effects to the Lognormal ($\ln X$ is Normal).
- The parent's **tail** sets the extreme family: exponential → Gumbel, polynomial → Fréchet, bounded → Weibull.

Don't confuse

- *"$\rho = 0$ means X and Y are independent."*
  → ρ measures linear dependence only: $\rho = 0 \not\Rightarrow$ independence.
- *"$4X_1$ has the same variance as a sum of four independent copies."*
  → Sum of four: $4\sigma^2$. But $\mathrm{Var}(4X_1) = 16\sigma^2$: no averaging out.
- *"More repetitions N shrink the spread of the sample mean."*
  → The variance σ^2/n shrinks with the sample size n, not with N.
- *"Design for the highest flood on record, with a margin."*
  → "Every flood on record was once worse than any flood on record." Design for the unobserved extreme.

Traps the questions test: 1. Uncertainty analysis

Foundations · Bayes

"A 99%-sensitive, 98%-specific spam filter flags an email, so it is almost surely spam."

→ With 1% spam, Pr(spam | flagged) = 0.0099/0.0297 = 1/3. The base rate matters (weld NDT: 0.43). (Tut1 Q7; Tut2 Q2)

Foundations · independence

"Mutually exclusive events are independent: $Pr(A \cap B) = Pr(A)Pr(B)$."

→ Mutually exclusive means $Pr(A \cap B) = 0$, so with Pr(A), Pr(B) > 0 they are dependent. (Quiz-1 Q1)

Functions of an rv

"For $Y = X^2$ with X uniform on (−1, 1), use $f_Y = f_X(\sqrt{y})|dx/dy|$."

→ $Y = X^2$ is many-to-one: add both roots, $f_Y(y) = 1/(2\sqrt{y})$ for $0 < y < 1$. (Quiz-2 Q1)

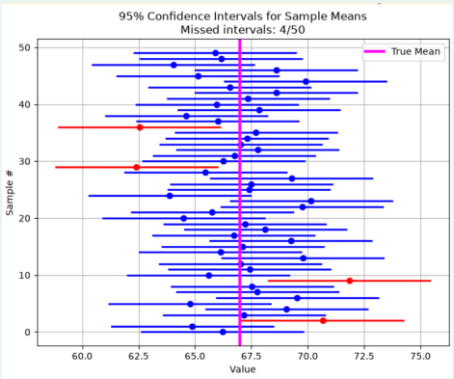
Multivariate · sums

"Var(D + L) = Var(D) + Var(L)."

→ Add $2\rho_{DL}\sigma_D\sigma_L$: $\sigma_T^2 = 100 + 225 + 120 = 445$ kN², $\sigma_T \approx 21.1$ kN. (Tut4 Q3)

2. Learning from data (Statistics): three parts

Question: *What do the data say?*

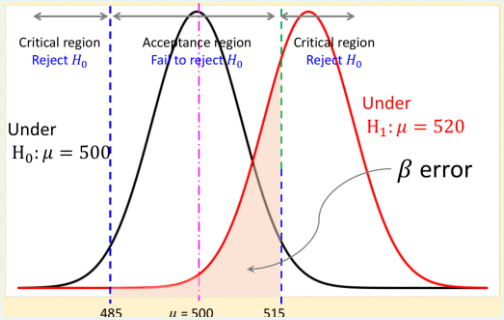


Estimate a number

M2C1 · L26–L27

- Parameters cannot be observed; a sample estimates them
- Bias and precision of an estimator
- Method of moments; maximum likelihood
- CI for μ: z if σ known, t if unknown; one-sided bounds
- CI for σ^2 via χ^2; choice of sample size

Stories: darts, Rome earthquakes, 4 of 50 misses, xkcd 882

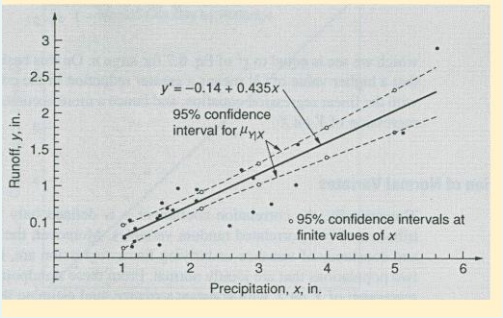


Test a claim

M2C2 · L-29–L-32

- H_0 and H_1; Type I (α) and Type II (β) errors
- Power = $1 - \beta$; p-value
- Procedure: hypotheses, statistic, tails, α, decision
- z- and t-tests on one and two means; pooled s_p
- χ^2 test on a variance

Stories: pregnancy test, justice system, airport security, Fe500 steel



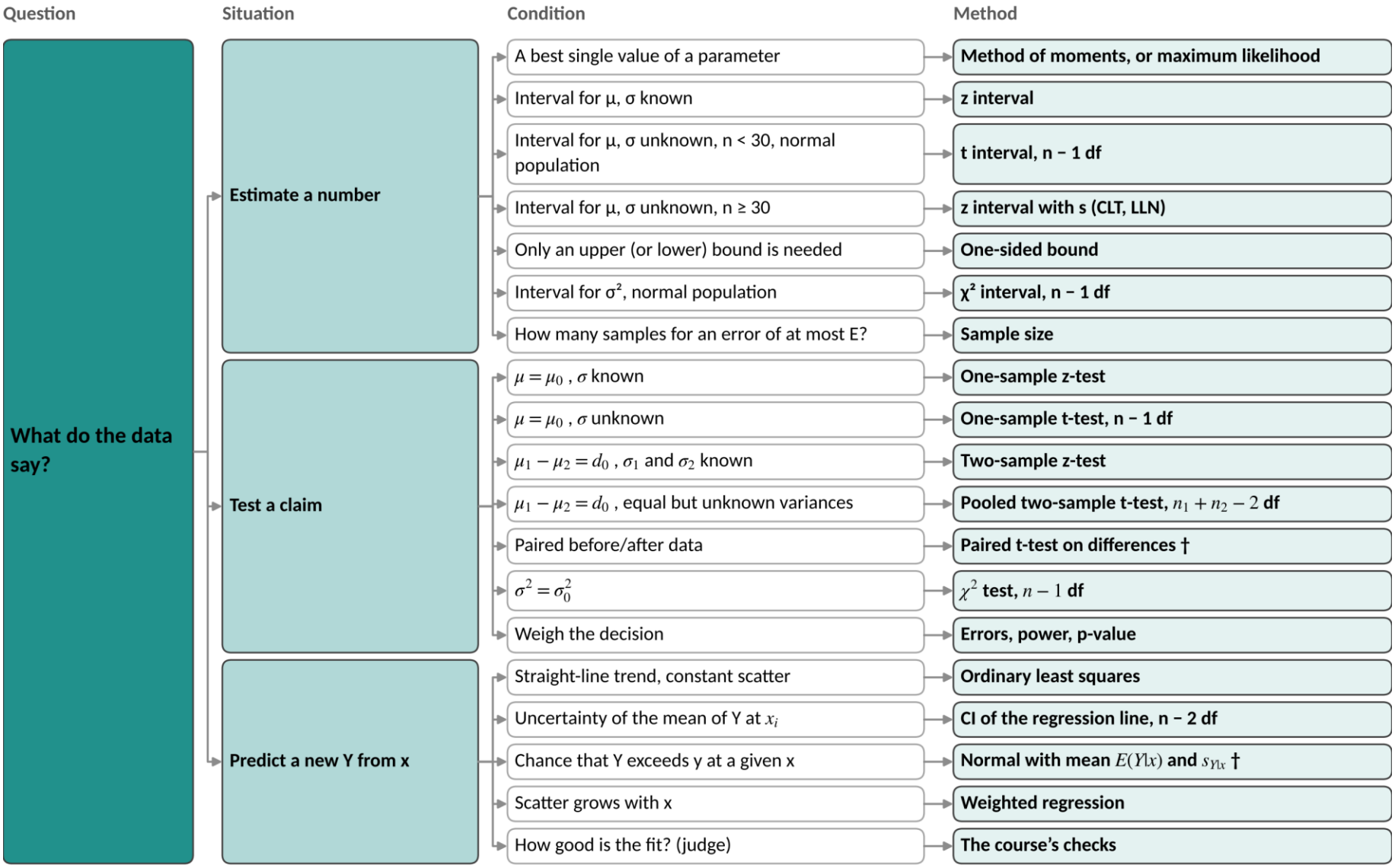
Predict a new Y

M2C3 · L-33–L-35

- Regression of Y on X; OLS estimates $\hat{\alpha}$, $\hat{\beta}$
- Conditional variance $s^2_{Y|x} = \Delta^2/(n-2)$
- CI of the regression line (t, $n - 2$ df)
- $R^2 = \hat{\rho}^2$ for one predictor
- Heteroscedasticity; weighted regression

Stories: 25 storms (rainfall vs. runoff), settlement data

Which tool when? 2. Learning from data



† Not printed on the lecture slides in this form. Paired data appear only as a table (8.pdf p49); the exceedance formula generalizes the worked example on 9.pdf p22.

Know this: 2. Learning from data (1 of 2)

Estimate a number (M2C1) · Test a claim (M2C2)

Know this

- **Method of moments** equates sample and distribution moments; **MLE** picks parameters that maximize the likelihood of the observed data.
- A point estimate says nothing about how close $\hat{\mu}$ is to μ; a **CI** adds this from the spread of $\bar{x}$, $\sigma/\sqrt{n}$.
- CI for μ: **z** if σ is known; **t** with $n-1$ df if σ is unknown and n < 30 (normal population). CI for σ^2: **χ^2**.
- α = Pr(reject H_0 | H_0 true) is the significance level; β = Pr(fail to reject | H_0 false); power = $1-\beta$.

Don't confuse

- *"There is a 95% probability that μ lies in this CI."*
  → Each CI contains μ or not. The 95% refers to the procedure.
- *"Failing to reject H_0 proves H_0."*
  → It is a "Not guilty" verdict. Proving H_0 "will take endless evidence."
- *"A Type I error is always the worse one."*
  → It depends on the stakes: for airport security or a bad batch, Type II is worse.
- *"The p-value is the probability that H_0 is true."*
  → It is how likely randomness alone gives a sample this extreme if H_0 is true.

Know this: 2. Learning from data (2 of 2)

Test a claim (M2C2) · Predict a new Y (M2C3)

Know this

- Test statistic: **z** if σ known; **t** ($n - 1$ df) if not; **pooled t** ($n_1 + n_2 - 2$ df) for two means with equal variances; **χ^2** for a variance.
- For a fixed n, α and β trade off; a larger n can reduce α.
- **OLS** minimizes the total squared error Δ^2; then $s^2_{Y|x} = \Delta^2/(n-2)$, since α and β are estimated from the data.
- For one predictor, the coefficient of determination is $R^2 = \hat{\rho}^2$.

Don't confuse

- *"The CI of the regression line has the same width everywhere."*
  → It widens away from $\bar{x}$: "farther from $\bar{x}$; bigger the error."
- *"OLS is fine even when the scatter grows with x."*
  → Weight the points: $w_i \propto 1/s^2_{Y|x}$.
- *"A strong correlation means one variable causes the other."*
  → Correlation does not imply causation (Miss America vs. steam murders).
- *"Darts close together means an accurate estimator."*
  → That is precision. Accuracy (bias) is closeness, on average, to the bullseye.

Traps the questions test: 2. Learning from data

Estimate · bias

"Dividing by n gives an unbiased sample variance."

→ $\mathrm{E}[\hat{\sigma}_1^{\,2}] = (\mathrm{n}-1)\sigma^2/\mathrm{n}$; dividing by $\mathrm{n}-1$ removes the bias. † (Tut6 Q2)

Estimate · sample size

"$n = (z_{\alpha/2}\sigma/E)^2 = 34.6$, so 34 cubes are enough."

→ Always round up: n = 35 for a half-width of at most 0.5 MPa. (Quiz-2 Q8c; Quiz-2 Q5)

Test · one-sided

"To test an upward shift at α = 0.05, compare z with 1.96."

→ A right-tailed test uses z_{α} = 1.645; $\mathrm{z}_{\alpha/2}$ = 1.96 is for two-sided tests. (Tut7 Q1, Q2)

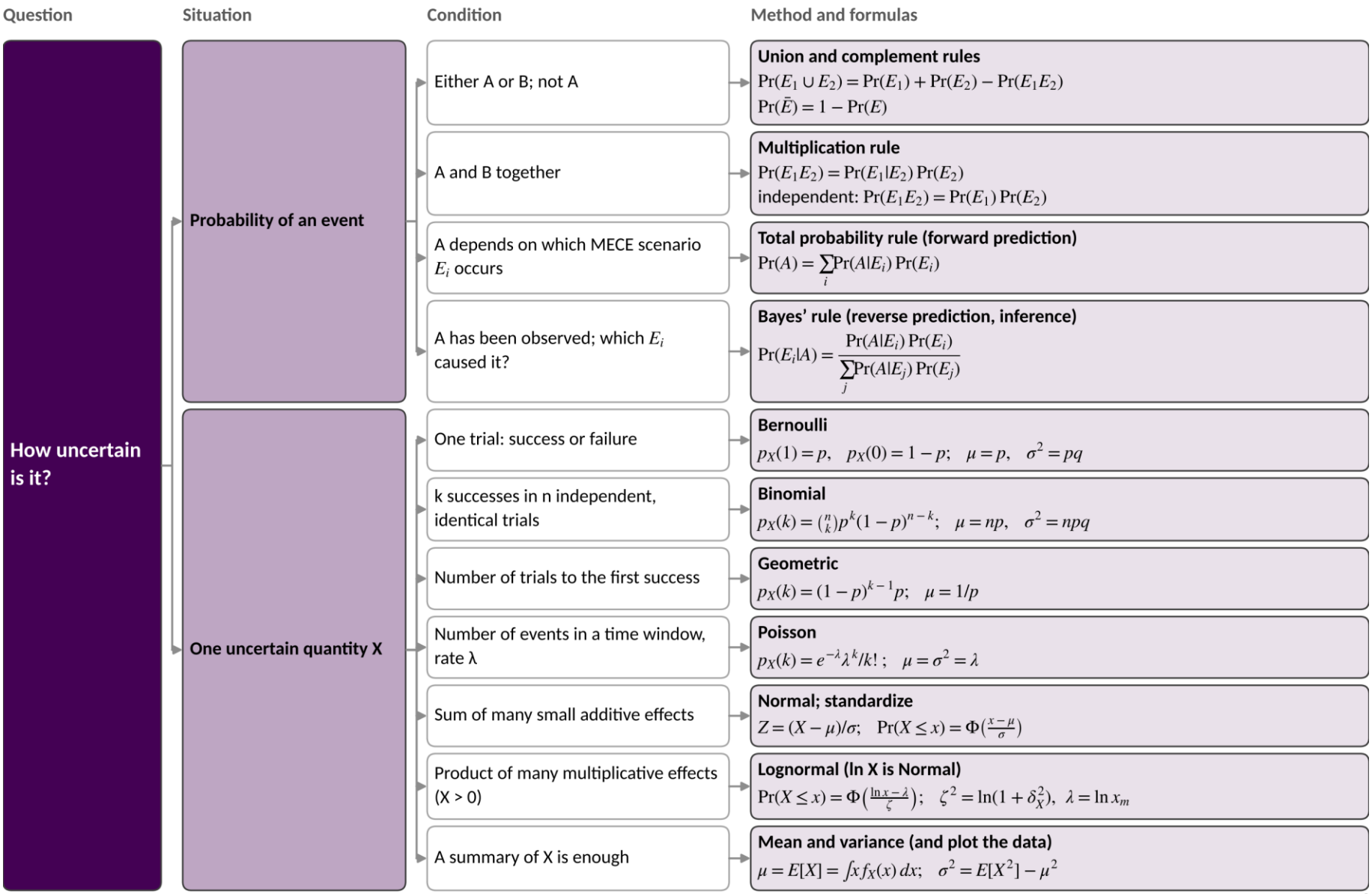
Predict · CI of the line

"The CI of the mean response is equally wide at every x."

→ It widens away from $\bar{\mathrm{x}}$: ±3.2 mm at $\bar{\mathrm{x}}$ = 22.53 kPa, ±10.6 mm at 150 kPa. (Tut8 Q3)

Thank you!

With formulas: 1. Uncertainty analysis (1 of 2)



With formulas: 1. Uncertainty analysis (2 of 2)

Question	Situation	Condition	Method and formulas
How uncertain is it?	Y = g(X) computed from uncertain inputs	g is linear: $Y = \sum a_i X_i + c$	**Exact propagation laws** $E[Y] = \sum a_i \mu_i + c$ $Var[Y] = \sum a_i^2 \sigma_i^2 + 2 \sum_{i<j} a_i a_j \, Cov[X_i, X_j]$
		One X, g monotonic, the PDF of Y is needed	**CDF method (Jacobian)** $f_Y(y) = f_X(g^{-1}(y)) \left\lvert \frac{dg^{-1}}{dy} \right\rvert$
		g nonlinear; mean and variance are enough	**Taylor approximation** $E[Y] \cong g(\mu_X) + \frac{1}{2}\left(\frac{\partial^2 g}{\partial x^2}\right)_{\mu_X} \sigma_X^2$ $Var[Y] \cong \left[\left(\frac{\partial g}{\partial x}\right)_{\mu_X}\right]^2 \sigma_X^2$
		g complex, or many inputs	**Monte Carlo simulation** $E[h(X)] \approx \frac{1}{N} \sum_{i=1}^{N} h(x_i)$
	Several quantities together	Joint outcomes; one variable given another	**Joint PDF, marginals, conditionals** $f_X(x) = \int f_{XY}(x, y)\, dy; \quad f_{Y\|X}(y\|x) = f_{XY}(x, y)/f_X(x)$
		Do they vary together (linearly)?	**Covariance and correlation** $\sigma_{XY} = E[XY] - E[X]E[Y]; \quad \rho_{XY} = \sigma_{XY}/(\sigma_X \sigma_Y)$ $\rho = 0 \not\Rightarrow$ statistical independence
	Averages and extremes	Mean of n i.i.d. values	**Central limit theorem** $\bar{X} \approx \mathcal{N}(\mu, \ \sigma^2/n)$
		Largest (smallest) of n values	**Exact EVD; asymptotic form set by the parent tail** $F_{Y_n}(y) = [F_X(y)]^n; \quad F_{Y_1}(y) = 1 - [1 - F_X(y)]^n$ exponential tail → Gumbel; polynomial → Fréchet; bounded → Weibull
		Rare events over a design life	**Return period; Poisson model** $T_r = 1/p_{f,1y}; \quad p_{f,T_d} = 1 - (1 - p_{f,1y})^{T_d}; \quad \Pr(\geq 1) = 1 - e^{-\lambda T}$

With formulas: 2. Learning from data (1 of 2)

Question	Situation	Condition	Method and formulas
What do the data say?	Estimate a number	A best single value of a parameter	**Method of moments, or maximum likelihood** MoM: set $E[X] = \bar{x}$, $Var[X] = s^2$ and solve MLE: $\hat{\theta} = \arg\max_{\theta} \ln \mathcal{L}(\theta\|\mathbf{x}), \quad \mathcal{L} = \prod_i f(x_i\|\theta)$
		Interval for μ, σ known	**z interval** $\bar{x} \pm z_{\alpha/2}\, \sigma/\sqrt{n}$
		Interval for μ, σ unknown, n < 30, normal population	**t interval, n − 1 df** $\bar{x} \pm t_{\alpha/2,\, n-1}\, s/\sqrt{n}$
		Interval for μ, σ unknown, n ≥ 30	**z interval with s (CLT, LLN)** $\bar{x} \pm z_{\alpha/2}\, s/\sqrt{n}$
		Only an upper (or lower) bound is needed	**One-sided bound** $\mu \le \bar{x} + z_{\alpha}\, \sigma/\sqrt{n}$
		Interval for σ^2, normal population	**χ^2 interval, n − 1 df** $\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}$
		How many samples for an error of at most E?	**Sample size** $n = \lceil (z_{\alpha/2}\, \sigma/E)^2 \rceil$
	Test a claim	$\mu = \mu_0$, σ known	**One-sample z-test** $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$
		$\mu = \mu_0$, σ unknown	**One-sample t-test, n − 1 df** $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
		$\mu_1 - \mu_2 = d_0$, σ_1 and σ_2 known	**Two-sample z-test** $z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$

With formulas: 2. Learning from data (2 of 2)

Question	Situation	Condition	Method and formulas
What do the data say?	**Test a claim (cont.)**	$\mu_1 - \mu_2 = d_0$, equal but unknown variances	**Pooled two-sample t-test,** $n_1 + n_2 - 2$ **df** $t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p\sqrt{1/n_1 + 1/n_2}}, \quad s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}$
		Paired before/after data	**Paired t-test on differences †** $d_i = x_i - y_i; \quad t = \frac{\bar{d}}{s_d/\sqrt{n}}, \quad n-1$ df
		$\sigma^2 = \sigma_0^2$	χ^2 **test,** $n-1$ **df** $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$
		Weigh the decision	**Errors, power, p-value** $\alpha = \Pr(\text{reject } H_0 \mid H_0 \text{ true}); \quad \beta = \Pr(\text{fail to reject } H_0 \mid H_0 \text{ false})$ power $= 1 - \beta$; two-tailed p-value $= 2[1 - \Phi(\lvert z_0 \rvert)]$
	Predict a new Y from x	Straight-line trend, constant scatter	**Ordinary least squares** $\hat{\beta} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}, \quad \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}; \quad s_{Y\lvert x}^2 = \frac{\Delta^2}{n-2}$
		Uncertainty of the mean of Y at x_i	**CI of the regression line, n − 2 df** $\bar{y}_i \pm t_{(1-\alpha/2),\,n-2}\; s_{Y\lvert x}\sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum(x_i - \bar{x})^2}}$
		Chance that Y exceeds y at a given x	**Normal with mean** $E(Y\lvert x)$ **and** $s_{Y\lvert x}$ **†** $\Pr(Y > y \mid x) = 1 - \Phi\left(\frac{y - E(Y\lvert x)}{s_{Y\lvert x}}\right)$
		Scatter grows with x	**Weighted regression** $\Delta^2 = \sum w_i[y_i - (\alpha + \beta x_i)]^2, \quad w_i \propto 1/s_{Y\lvert x_i}^2$ (e.g., $1/x_i^2$)
		How good is the fit? (judge)	**The course's checks** $R^2 = \hat{\rho}^2, \quad \hat{\rho} = \hat{\beta}\, s_X/s_Y$; compare $s_{Y\lvert x}$ with s_Y CI bands widen away from $\bar{x}$; look for scatter that grows with x

† Not printed on the lecture slides in this form. Paired data appear only as a table (8.pdf p49); the exceedance formula generalizes the worked example on 9.pdf p22.

Core ideas: probability

Rules and random variables

- Events combine by a few rules: complements, unions, conditioning, independence
- Conditioning changes probabilities; Bayes' rule updates them as evidence arrives, and base rates matter
- A random variable has a distribution; mean, spread, skew and tails each tell you something different
- Dependence between variables changes the risk of combinations

Limits, extremes and propagation

- Law of large numbers: averages settle down as samples grow
- Central limit theorem: sums of many small effects look Normal
- Maxima follow their own laws; exposure time matters ("a 100-year flood" can occur twice in a decade)
- Uncertainty in inputs flows through models; linearize when it is small, simulate when it is not

Core ideas: statistics

Estimating and testing

- A sample stands in for a population; how it was drawn matters more than its size
- Every estimate needs a standard error; trade bias against variance
- A confidence interval is a procedure that covers the truth at a stated rate
- A test weighs two errors (false alarm, missed effect); power is the chance of seeing a real effect

Modeling and checking

- Regression describes how Y varies with x, including the scatter around the line
- Judge a model on data it was not fitted to (hold-out validation, cross-validation)
- Check residuals and fit with plots, not only with a single number
- “All models are wrong, but some are useful” (George Box)

Experts disagree: Should fixed thresholds such as $p < 0.05$ decide what counts as a finding? Some statisticians argue for reporting effect sizes with intervals and dropping the thresholds; others keep them as a practical rule for decisions. Bayesian analysts report posterior probabilities instead.